

AW: Methodology question — bundle selection in hierarchical prefetching (ASPLOS '25)

Probst, Benjamin-Elias

Mi 13.05.2026 23:37

Gesendete Elemente

An:ztj22 <ztj22@mails.tsinghua.edu.cn>;

Cc:Habich, Dirk <dirk.habich@tu-dresden.de>; zyh02@tsinghua.edu.cn <zyh02@tsinghua.edu.cn>; Boris Grot <Boris.Grot@ed.ac.uk>;

Dear Tingji,

Thank you, that helped me a lot!

Mit freundlichen Grüßen,
Benjamin-Elias Probst

Von: ztj22 <ztj22@mails.tsinghua.edu.cn>

Gesendet: Montag, 11. Mai 2026 14:16:38

An: Probst, Benjamin-Elias

Cc: Habich, Dirk; zyh02@tsinghua.edu.cn; Boris Grot

Betreff: Re: Methodology question — bundle selection in hierarchical prefetching (ASPLOS '25)

Dear Benjamin,

Thank you for your interest in our work and for your kind words.

I am attaching a small demo package that may help with your implementation. The zip file contains a Python script and a README. It provides an example processing script explaining the call graph analysis process used in Hierarchical Prefetching.

It should capture the core bundle-selection logic and hopefully serve as a useful reference for adapting the idea to your PRT-ART implementation.

Please feel free to let us know if anything is unclear.

Best regards,

Tingji

----- Replied Message -----

From [Boris Grot<Boris.Grot@ed.ac.uk>](mailto:Boris.Grot@ed.ac.uk)

Date 5/10/2026 21:36

To [ztj22<ztj22@mails.tsinghua.edu.cn>](mailto:ztj22@mails.tsinghua.edu.cn)

Cc [Dirk Habich<dirk.habich@tu-dresden.de>](mailto:Dirk.Habich@tu-dresden.de),
[Benjamin-Elias Probst<benjamin-elias.probst@mailbox.tu-dresden.de>](mailto:Benjamin-Elias.Probst@mailbox.tu-dresden.de),
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Subject Re: Methodology question — bundle selection in hierarchical prefetching (ASPLOS '25)

Hi Tingjie,

Do you think you could help with the query below. Thanks!

-Boris

From: Benjamin-Elias Probst <benjamin-elias.probst@mailbox.tu-dresden.de>
Date: Friday, 8 May 2026 at 18:59
To: zyh02@tsinghua.edu.cn <zyh02@tsinghua.edu.cn>, Boris Grot <Boris.Grot@ed.ac.uk>
Cc: Dirk Habich <dirk.habich@tu-dresden.de>
Subject: Methodology question — bundle selection in hierarchical prefetching (ASPLOS '25)

Sie erhalten nicht häufig E-Mails von benjamin-elias.probst@mailbox.tu-dresden.de. [Erfahren Sie, warum dies wichtig ist](#)

Dear Prof. Zhang, Dear Prof. Grot,

I am working on my diploma thesis at TU Dresden, supervised by Prof. Dr. Dirk Habich. The thesis title is "Active Cache-Aware Hardware Adaptation Cache Engine for Trie-Based Index Structures".

PUBLICATION OF INTEREST

Title: "Hierarchical Prefetching: A Software-Hardware Instruction Prefetcher for Server Applications"
Authors: Tingji Zhang, Boris Grot, Wenjian He, Yashuai Lv, Peng Qu*, Fang Su, Wenxin Wang, Guowei Zhang, Xuefeng Zhang, Youhui Zhang*
Venue: ASPLOS '25, 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2
DOI: 10.1145/3676641.3716260
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Your bundle mechanic for hierarchical prefetching is a direct conceptual inspiration for one of the prefetch strategies in my cache engine — it sits on axis 7 (Prefetch) of our building-block cross-permutation matrix.

I have already located your repository CRAFT-THU/gem5-hp (BSD-3-Clause), which is a gem5 fork. Since my work focuses on CPU-only database index structures (not on instruction prefetching), I do not need the full gem5 simulator but would like to transfer the bundle concept to a software-prefetch strategy on data structures.

Question: Is there an isolated description of the bundle-selection algorithm (for example as pseudocode, a short code snippet, or an isolated module outside the gem5 integration) that I may use as the basis for a PRT-ART-internal implementation via an adapter? A short technical note or a pointer to a self-contained reference implementation would be very helpful.

I will of course cite your work and preserve the BSD-3-Clause license on any derived components.

Best regards,
Benjamin-Elias Probst

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